

Target and Precursor named entities recognition from scientific texts of high-temperature steel using deep neural network

M. Saef Ullah Miah^{1,2}[0000-0003-4587-4636], Junaida Sulaiman²[0000-0002-7834-5338], Talha Bin Sarwar^{1,2}[0000-0001-5974-1282], Imam Ul Ferdous²[0000-0002-9158-216X], Saima Sharleen Islam³[0000-0002-3891-4460], and Md. Samiul Haque⁴[0000-0002-3333-6868]

¹ American International University-Bangladesh, Dhaka, Bangladesh

² Universiti Malaysia Pahang, Pekan, Malaysia

³ University of Trento, Trento, Italy

⁴ University of Dhaka, Dhaka, Bangladesh

Abstract. Named Entity Recognition (NER) is an essential task in natural language processing, especially in the domain of scientific texts. This paper presents a study of NER for scientific texts in high-temperature steel, a type of alloy used in various applications where high temperatures prevail. We propose a NER system using Bi-LSTM with a domain-specific embedding approach and evaluate its performance on a test dataset. The study results show that the proposed NER system achieves an F1 score of 0.99, indicating that it can accurately identify and classify named entities in scientific texts about high-temperature steel with high precision and recall. The proposed approach was more effective than the classical machine learning-based approach. Our results suggest that the domain-specific embedded Bi-LSTM technique can be an effective approach for NER in scientific texts, especially in specialized domains such as high-temperature steel.

Keywords: Material Named Entity Recognition · High-temperature steel · Bi-LSTM · Deep learning in industry · Material Natural Language Processing

1 Introduction

Named entity recognition (NER) is a critical task in natural language processing, with applications ranging from information extraction and text summarization to machine translation and question answering. In the scientific text, NER is particularly important for extracting and organizing information about specific concepts, techniques, and tools, often mentioned using technical terms and abbreviations.

High-temperature steel [9] is a type of alloy steel used in various applications where high temperatures are encountered, such as power generation, aerospace, and automotive engineering. Accurate identification and classification of named

entities in the scientific text on high-temperature steel are important for tasks such as information extraction and text summarization and for enabling the automated analysis of scientific literature in this domain.

However, NER in scientific texts can be challenging due to the named entities’ complexity and the lack of annotated data for training and evaluating NER systems [14]. In this paper, we present a study of NER for scientific texts in the domain of high-temperature steel, focusing on the key challenges and approaches to this task, as well as the recent advances and areas for future research. We also propose a NER system that utilizes Bi-LSTM [4] with a domain-specific embedding approach and evaluates its performance on a domain expert-curated test dataset [10].

To conduct the study, we have reviewed the existing literature on NER for scientific texts in the domain of high-temperature steel, using a variety of databases such as the ACM Digital Library, IEEE Xplore, and PubMed. We have analyzed the methods used for identifying and classifying named entities, the performance of different NER systems, and the challenges and limitations of these systems. Our study has identified several key challenges in NER for scientific texts in high-temperature steel. These include the complexity of the named entities, the lack of annotated data, the ability to handle synonyms and variations of named entities, and named entities mentioned in different ways in the text. To address these challenges, we propose the NER system, which utilizes a domain-specific embedding with the Bi-LSTM technique.

We evaluate the performance of the proposed NER system on a test dataset and compare it to several classic machine learning-only approaches [7]. Pre-trained models fine-tuned with the test dataset, namely BERT [3] and sciBERT [1].

2 Related Works

Kim et al. investigate the use of neural networks in predicting synthesis conditions for inorganic materials [8]. The Conditional Variational Autoencoder (CVAE) model identifies influential parameters and suggests changes for optimal results. It learns from literature data and generalizes to new materials, enabling diverse applications. The model achieves accuracy of 59%, 67%, and 91% on training datasets with 100, 1000, and 10,000 precursors, respectively. Increasing latent dimensions to 64 improves accuracy to 98%, but affects conditioning similarly to the synthesis action CVAE.

Cruse et al. present a dataset extracted from the scientific literature on gold nanoparticle synthesis [2]. The text-mining algorithms accurately capture information on synthesis methodologies, nanoparticle morphologies, and sizes. The dataset enables the study of synthesis-process associations and facilitates the design of new materials.

Wang et al. propose an automated NLP pipeline for superalloy data, extracting chemical composition and property data from 14425 journal articles [16]. Their synthesis and characterization of three alloys confirm the pipeline’s accu-

racy. The recovered database through text mining provides a valuable superalloy production and global analysis resource.

Gupta et al. compares MatSciBERT to SciBERT, showing MatSciBERT’s superior performance in recognizing named entities, relation classification, and abstract classification [5]. MatSciBERT achieves an 82% Macro-F1 score on the SOFC-Slot dataset using five cross-validation splits.

3 Methodology

Several steps are performed to create the proposed deep neural network model. First, pertinent scientific publications are collected in PDF format and converted to plain text with appropriate sentence boundaries [11]. The dataset is then annotated and validated by domain experts. The dataset is produced in IOB format. After preparing the dataset, the model TP-NER is created, trained, tested, and fine-tuned to ensure it reliably recognizes named entities in unseen text. Finally, the proposed approach extracts named entities from scientific articles.

3.1 Data collection and processing

Through the electronic resource of Universiti Malaysia Pahang’s library, scientific papers are downloaded [15]. Using the search term “high temperature steel AND P91 P92”, the top 500 most-cited scholarly articles are collected and analyzed for use in this investigation.

3.2 Dataset preparation for TP-NER model training

Experts in the domain have used 50 relevant papers to compile the dataset. 1551 sentences from these articles are annotated with the labels B-target, I-target, B-precursor, I-precursor, and O. The output of the annotation process is saved in IOB format. The annotated dataset contains 426 tokens with the target class, which combines B-target and I-target labels, and 486 tokens with the precursor class, which combines B-precursor and I-precursor labels. The inter-annotator agreement (IAA) score for this dataset is 94% of Cohen’s Kappa, indicating a high level of agreement between the annotators as they assigned labels to the texts. The dataset can be accessed via Mendeley data [10].

3.3 TP-NER: deep neural network model development

There are 5 layers in the proposed model, with 3 of them being concealed. The first layer is an input layer, the second is a domain-specific integer-coded word embedding layer, the third is a dropout layer, the fourth is a Bi-LSTM layer with TanH activation and 200 neurons, and the fifth and final layer is a time-distributed dense layer with softmax activation. Algorithm 1 illustrates the architecture and layer-by-layer setup.

Algorithm 1 Architecture of the Proposed TP-NER DNN model

```

1: Input: Sequence with token and labels: SL
2: num_words = unique words in dataset, num_tags = unique labels in dataset
3: set maxLen = 60
4: For {S in sent_list} { X = pad sequences words }
5: For {S in sent_list} { y = pad sequences labels, y = hot encode y }
6: Split data into training and testing set
7: input_word = (maxLen)
8: Embedding(input_dim = num_words, output_dim = maxLen, input_length =
   maxLen)
9: SpatialDropout1D(0.2)
10: Bidirectional(LSTM(units = 200, return_sequences = True, recurrent_dropout =
   0.2))
11: out = TimeDistributed(Dense(activation = 'softmax'))
12: matrec = Model(input_word, out)
13: matrec.compile(optimizer = 'adam', loss = 'categorical_crossentropy', metrics =
   [accuracy, precision_mt, recall_mt, f1_mt])
14: matrec.fit(train-test data, batch_size = 16, verbose = 1, epochs = 50,
15: validation_split = 0.2, callbacks = [tensorboard_cbk, es])
16: model.save('tpner.h5')
17: Return medner.h5

```

3.4 Evaluation metrics

The proposed TP-NER model is evaluated utilizing Precision(*Pr*), Recall(*Rcl*), and *F1* metrics. If each token is identified correctly during entity prediction, the entity is marked as true. True Positives (*TPs*) are calculated when the entities are accurately predicted. However, False Positives (*FPS*) are indicated when the predicted beginning token does not match the marked token of the entity. False Negatives (*FNg*) are recorded whenever the system wrongly forecasts the initial token of expected entities. The equations in equation (1) calculate the evaluation metrics.

$$Pr = \frac{TPs}{TPs + FPS}, Rcl = \frac{TPs}{TPs + FNg}, F1 = \frac{2 * Pr * Rcl}{Pr + Rcl} \quad (1)$$

4 Result and Discussion

This study examines the efficacy of machine learning methods for recognising target and precursor-named entities for high-temperature steel from scientific publications. On a dataset of published scientific publications, the study compares the performance of numerous algorithms, including BERT [3], sci-BERT [1], Decision Tree [13], Random Forest [6], and Gaussian Naive Bayes [12]. Table 1 presents the outcomes of the experiments. The proposed TP-NER model based on Bi-LSTM achieves the highest *F1* score of 0.99, followed by Decision Tree with an *F1* score of 0.92 and Random Forest with an *F1* score of 0.91. In addition, the performance of the approaches improves when the training dataset is domain-specific. Still, larger BERT and sci-BERT models perform less adequately despite being fine-tuned using the same dataset. This observation presents an opportunity to explore larger language models with a growing volume

Table 1. Precision, recall and F1 scores for different models

Model Name	Precision	Recall	F1
Proposed TP-NER	0.9972	0.9964	0.9968
BERT	0.81	0.865	0.8365
sci-BERT	0.88	0.81	0.85
Decision Tree	.89	.95	.923
Random Forest	.91	0.92	.91
Gaussian Naive Bayes	.85	.88	.86

of domain-specific data to determine whether their performance may be enhanced. In contrast, the experimental results demonstrate that a domain-specific embedded neural network model without pre-trained embedding and language models outperforms finely tuned large pre-trained models for target and precursor-named entity recognition in high-temperature steel-related scientific papers.

The experimental results show that the proposed model for target and precursor named entity recognition from scientific text is the most successful algorithm among those compared. However, an additional study is required to evaluate the effectiveness of the suggested model on various forms of texts, such as patent records and technical reports.

5 Conclusion and Future work

This study demonstrates the effectiveness of the proposed deep learning-based method for recognising target and precursor named entities. The experimental results show that the proposed method outperforms conventional machine learning techniques and pre-trained models and achieves high precision, recall, and F1 score of 0.9972, 0.9964, and 0.9968, respectively, in recognizing different types of named entities related to high-temperature steel, such as targets and precursors.

To further validate the proposed deep learning model, it is planned to extend the study to other material domains and languages in the future. In addition, the proposed model can combine the named entity identification system with different natural language processing approaches to improve the system’s overall performance.

Acknowledgment

This research was supported by the Post Graduate Research Scheme of Universiti Malaysia Pahang entitled: An automated knowledge-based framework for EDLC using deep learning approach, PGRS220337.

References

1. Beltagy, I., Lo, K., Cohan, A.: Scibert: A pretrained language model for scientific text. arXiv preprint arXiv:1903.10676 (2019)
2. Cruse, K., Trewartha, A., Lee, S., Wang, Z., Huo, H., He, T., Kononova, O., Jain, A., Ceder, G.: Text-mined dataset of gold nanoparticle synthesis procedures, morphologies, and size entities. Scientific Data **9**(1), 234 (2022)

3. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: BERT: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers). pp. 4171–4186. Association for Computational Linguistics, Minneapolis, Minnesota (Jun 2019). <https://doi.org/10.18653/v1/N19-1423>, <https://aclanthology.org/N19-1423>
4. Graves, A., Schmidhuber, J.: Framewise phoneme classification with bidirectional lstm and other neural network architectures. *Neural networks* **18**(5-6), 602–610 (2005)
5. Gupta, T., Zaki, M., Krishnan, N.A.: Matscibert: A materials domain language model for text mining and information extraction. *npj Computational Materials* **8**(1), 102 (2022)
6. Ho, T.K.: Random decision forests. In: Proceedings of 3rd international conference on document analysis and recognition. vol. 1, pp. 278–282 (1995)
7. Islam, S.S., Haque, M.S., Miah, M.S.U., Sarwar, T.B., Nugraha, R.: Application of machine learning algorithms to predict the thyroid disease risk: an experimental comparative study. *PeerJ Computer Science* **8**, e898 (2022)
8. Kim, E., Jensen, Z., van Grootel, A., Huang, K., Staib, M., Mysore, S., Chang, H.S., Strubell, E., McCallum, A., Jegelka, S., et al.: Inorganic materials synthesis planning with literature-trained neural networks. *Journal of chemical information and modeling* **60**(3), 1194–1201 (2020)
9. Meetham, G.W.: High-temperature materials — a general review. *Journal of Materials Science* **26**(4), 853–860 (Feb 1991). <https://doi.org/10.1007/BF00576759>, <http://link.springer.com/10.1007/BF00576759>
10. Miah, M.S.U., Sulaiman, J., FERDOUS, I.U., Sarwar, T.B.: Tp-ner: A named entity recognition dataset of target and precursor named entities for high-temperature steel (2022). <https://doi.org/10.17632/5ZNG6KHY9H.1>, <https://data.mendeley.com/datasets/5zng6khy9h/1>
11. Miah, M.S.U., Sulaiman, J., Sarwar, T.B., Naseer, A., Ashraf, F., Zamli, K.Z., Jose, R.: Sentence boundary extraction from scientific literature of electric double layer capacitor domain: tools and techniques. *Applied Sciences* **12**(3), 1352 (2022)
12. Rish, I., et al.: An empirical study of the naive bayes classifier. In: IJCAI 2001 workshop on empirical methods in artificial intelligence. vol. 3, pp. 41–46 (2001)
13. Rokach, L.: Decision forest: Twenty years of research. *Information Fusion* **27**, 111–125 (2016)
14. Sharma, A., Amrita, Chakraborty, S., Kumar, S.: Named Entity Recognition in Natural Language Processing: A Systematic Review. In: Gupta, D., Khanna, A., Kansal, V., Fortino, G., Hassanien, A.E. (eds.) Proceedings of Second Doctoral Symposium on Computational Intelligence, vol. 1374, pp. 817–828. Springer Singapore, Singapore (2022). https://doi.org/10.1007/978-981-16-3346-1_66, https://link.springer.com/10.1007/978-981-16-3346-1_66
15. UMP Library: e-Resource UMP Lib. <https://login.ezproxy.ump.edu.my/login> (2022), <https://login.ezproxy.ump.edu.my/login>
16. Wang, W., Jiang, X., Tian, S., Liu, P., Dang, D., Su, Y., Lookman, T., Xie, J.: Automated pipeline for superalloy data by text mining. *npj Computational Materials* **8**(1), 9 (2022)